CAMBRIDGE GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

0775 Further Mathematics 1

JUNE 2020

ADVANCED LEVEL

Centre Number	
Centre Name	
Candidate Identification No.	
Candidate Name	

Mobile phones are NOT allowed in the examination room

MULTIPLE CHOICE QUESTION PAPER

One and a half hours

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.

2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins:

3. Check that this question booklet is headed **Advanced Level – 0775 Further Mathematics 1.**
4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided on the answer sheet using your HB pencil: **Candidate Name, Exam Session, Subject Code and Candidate Identification Number.** Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.

How to answer the questions in this examination

6. Answer **ALL** the **50** questions in this Examination. All questions carry equal marks.
7. Calculators are allowed.
8. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide which answer is appropriate. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square bracket for the answer you have chosen.

For example, if **C** is your correct answer, mark **C** as shown below:

[A] [B] [C] [D]

9. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to this question later.
11. Do all rough work in this booklet using the blank spaces in the question booklet.
12. **At the end of the examination, the invigilator shall collect the answer sheet first and then the question booklet. DO NOT ATTEMPT TO LEAVET THE EXAMINATION HALL WITH IT.**

Turn over

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1. The integrating factor of the differential equation $\frac{dy}{dx} + \frac{y}{2x} = \frac{e^x}{\sqrt{x}}$, $x \neq 0$, is

A $\frac{1}{2}x$

B x^2

C $\frac{1}{x^2}$

D $2x$

2. $\frac{1}{e^{2i\theta}} \equiv$

A $e^{-2}(\cos\theta + i\sin\theta)$

B $e(\cos 2\theta + i\sin 2\theta)$

C $\cos 2\theta - i\sin 2\theta$

D $e^{-2}(\cos\theta - i\sin\theta)$

3. The expression $\frac{x^2}{(x^2+2)^2}$ in partial fractions, where P, Q, R, S are real constants, is

A $\frac{P}{x^2+2} + \frac{Q}{(x^2+2)^2}$

B $\frac{Px+Q}{x^2+2} + \frac{Rx+S}{(x^2+2)^2}$

C $\frac{Px+Q}{x^2+2} + \frac{Rx^2+S}{(x^2+2)^2}$

D $\frac{P}{x^2+2} + \frac{Qx+R}{(x^2+2)^2}$

4. The series $\sum_{r=1}^{\infty} r^{-\frac{k}{2}}$ is divergent for

A $k > 1$

B $k \geq 2$

C $k \leq 2$

D $0 < k \leq 2$

5. The image of the vector $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ under a mapping defined by the matrix M, where $M = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$ is

A $-3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$

B $3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$

C $-3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$

D $3\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}$

6. The curve C has parametric equations $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $a > 0$, $0 \leq t \leq 2\pi$. The area of the region bounded by the curve and the x-axis is $3\pi a^2$. The mean value of y with respect to x for $0 \leq t \leq 2\pi$ is

A $\frac{3a^2}{2}$

B $\frac{3a}{2}$

C $\frac{3}{2}$

D $\frac{3\sqrt{a}}{2}$

7. A tangent at the pole to the polar curve $r = \sqrt{2} - 2\cos\theta$ is $\theta =$

A $\frac{\pi}{6}$

B $\frac{\pi}{4}$

C $\frac{\pi}{3}$

D $\frac{\pi}{2}$

8. Which one of the following conics is a parabola

A $x^2 + y + 5 = 4x$

B $2x^2 + y^2 = 4x + 4$

C $4x^2 - 9y^2 + 32x - 144y = 548$

D $25(x+2)^2 - 36(y-1)^2 = 900$

9. If the series $\sum_{r=1}^{\infty} \frac{3}{r^{k+1}}$ converges, then

A $k > -1$

B $k > 0$

C $k > 2$

D $k > 3$

10. Given the group $(S = \{a, b, c, d\}, \otimes)$

$\otimes$	a	b	c	d
a	d	c	b	a
b	c	d	a	b
c	b	a	d	c
d	a	b	c	d

The inverse of an element x, $x \in S$ is

A x

B d

C c

D b

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11. A continuous random variable X has a probability density function f, where

$$f(x) = \begin{cases} k(1-x^2) & 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}.$$

The cumulative probability function $F(x) =$

A $\begin{cases} 0, & x < 0 \\ k(1-x^2), & 0 < x \le 1 \\ 1, & x > 1 \end{cases}$

B $\begin{cases} k(x - \frac{x^3}{3}), & 0 < x \le 1 \\ 0, & \text{elsewhere} \end{cases}$

C $\begin{cases} 0, & x < 0 \\ k(x - \frac{x^3}{3}), & 0 < x \le 1 \\ 1, & x > 1 \end{cases}$

D $\begin{cases} 0, & x < 0 \\ k(x - \frac{x^3}{3}), & 0 < x \le 1 \\ 0, & x > 1 \end{cases}$

12. A particle of mass m falls against a resistance of magnitude kv, where vis the velocity and k is a positive constant. The equation of motion is given by

A $\frac{dv}{dt} = -mkv$

B $\frac{dv}{dt} = mg - kv$

C $\frac{dv}{dt} = mg + kv$

D $\frac{dv}{dt} = -mg - kv$

13. If x is the displacement of a particle from a fixed point O at time t, then which of the following represents oscillatory motion?

A $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 0$

B $\frac{d^2x}{dt^2} - 4\frac{dx}{dt} - 5x = 0$

C $\frac{d^2x}{dt^2} + 4x = 0$

D $\frac{d^2x}{dt^2} - 4x = 0$

14. If $f(x) = 2 + x^2 - x^3$, then the equation $f(x) = 0$ has a solution on the interval

A $[1,2]$

B $[0,1]$

C $[-1,0]$

D $[-2,-1]$

15. Given that $f(x) = \begin{cases} \frac{2\sin(\frac{x}{2})}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$

is a continuous function at $x = 0$, the value of k is

A 0

B $\frac{1}{2}$

C 1

D 2

16. A force $\mathbf{F} = (3\mathbf{i} - \mathbf{j} + 2\mathbf{k})N$ acts on a particle giving it a displacement of $(3\mathbf{i} + 4\mathbf{k})$m. The work done by $\mathbf{F}$ is

A 15 J

B $5\sqrt{14}$ J

C 5 J

D 17 J

17. The centre of symmetry of the curve $y = \frac{2x}{x^2 - 4}$, $x \neq \pm 2$ is

A $(0,2)$

B $(2,0)$

C $(0,0)$

D $(-2,1)$

18. Given the truth table of a conjunction, then the values of a and b respectively are:

P	Q	P ∧ Q
T	T	T
T	F	a
F	T	F
F	F	b

A F,T

B T,F

C T,T

D F,F

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19. If two elastic spheres collide and the total mechanical energy remains constant, then the value of the coefficient of restitution e is

A $0 < e \leq 1$
B $0 \leq e < 1$
C $e = 0$
D $e = 1$

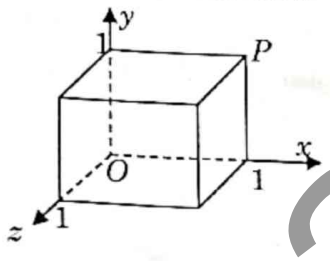
20. $\int_0^4 [x]dx =$, where $[x]$ is the greatest integer function of x

A 0
B 4
C 6
D 8

21. The number of solution(s) of the equation $3\cosh(2x-1) = 3$ is

A 0
B 1
C 2
D 3

22. The coordinates of the point P on the unit cube is



A (1,1,0)
B (0,1,1)
C (1,0,1)
D (1,1,1)

23. Given that $f(x) = \begin{cases} 2\cosh x, & x < 0 \\ \sinh x, & x \geq 0. \end{cases}$

Then the $\lim_{x \to 0} f(x)$

A does not exist
B is 2
C is 1
D is 0

24. The depth of the cross-section of a river flowing at 2 m/s from riverbank to bank measures

distance across, x	0	1	2
depth, y	0.5	1.2	0.4

A Simpson's rule approximation of the volume of water released each second is

A 7.6 m^3/s
B 3.8 m^3/s
C 4.0 m^3/s
D 2.5 m^3/s

25. A parabola is the locus of a point equidistant from

A two fixed lines
B two fixed points
C a fixed point and a fixed line
D two fixed perpendicular lines

26. If $\cosh 2x = \frac{5}{4}$ then the positive value of $\sinh 2x$ is

A $\frac{4}{5}$
B $\frac{3}{2}$
C $\frac{3}{4}$
D $\frac{4}{3}$

27. A medical treatment has a success rate of 90%. Two patients are administered the treatment independently. The probability that neither of them gets well is

A 0.90
B 0.81
C 0.10
D 0.01

28. The moment of inertia of a body of mass $3m$ abou an axis is $9ma^2$. The radius of gyration of the bod about the given axis is

A $3a$
B $a\sqrt{3}$
C $\frac{1}{3}a$
D $a\frac{\sqrt{3}}{3}$

29. The unit digit of the number $19^{2004} + 11^{2004} - 9^{2}$ is

A 1
B 3
C 7
D 9

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30. The vertices A, B,C of ΔABC have position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively. The area of ΔABC is

A $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$

B $\frac{1}{2}|\mathbf{a} \times \mathbf{c}|$

C $\frac{1}{2}|\mathbf{a} \times \mathbf{b} \times \mathbf{c}|$

D $\frac{1}{2}|(\mathbf{a} - \mathbf{b}) \times (\mathbf{b} - \mathbf{c})|$

---

31. A particle is moving round the polar curve $r = 1 + \sin\theta$ with constant angular velocity w. The maximum speed of the particle is

A w

B $w\sqrt{2}$

C $w / 2$

D $2w$

---

32. If the function $f(x) = \begin{cases} \dfrac{x^2 - 1}{x + 1}, & x \neq -1 \\ k & x = -1 \end{cases}$

is continuous on $\mathbb{R}$, then $k =$

A -2

B -1

C 0

D 1

---

33. If $z = \left[i\left(\cos\theta + i\sin\theta\right)\right]^n$ then $\arg(z) =$

A $-2n\theta + n\frac{\pi}{2}$

B $n\theta + n\frac{\pi}{2}$

C $2n\theta$

D $2n\theta - n\frac{\pi}{2}$

---

34. The particular integral of the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x}$ is

A λe^{2x}

B $\lambda x e^{2x}$

C $\lambda x^2 e^{2x}$

D $2\lambda x e^{2x}$

---

35. Given that $f(x)$ is a continuous even function, then $\int_{-a}^{a} f(x)dx =$

A 0

B $2\int_{0}^{a} f(x)dx$

C $2\int_{-a}^{a} f(x)dx$

D $\int_{0}^{a} f(x)dx$

---

36. Given that $\frac{dy}{dx} + 2y = 0$ and $y = 1$ when $x = 0$. A quadratic approximation for y is

A $1 - 2x - 2x^2$

B $1 - 2x + 2x^2$

C $1 - 2x + x^2$

D $1 - 2x + \frac{1}{2}x^2$

---

37. The function f has a removable discontinuity at x_0. Therefore

A $f(x_0)$ exists

B $\lim_{x \to x_0} f(x)$ exists

C $\lim_{x \to x_0} f(x) = f(x_0)$

D $\lim_{x \to x_0} f(x)$ does not exist

---

38. If $x > 0$, then $\frac{d}{dx}(\ln\sinh 2x) =$

A $\coth 2x$

B $2\coth 2x$

C $2\tanh 2x$

D $\tanh 2x$

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39. $\frac{x^2}{x^2-1}$ expressed in partial fractions, where P, Q, and R, are real constants, with $x \neq \pm 1$, is

A $\frac{P}{x+1}+\frac{Q}{x-1}$

B $P+\frac{Q}{x+1}+\frac{R}{x-1}$

C $\frac{P}{x-1}+\frac{Q}{(x-1)^2}$

D $P+\frac{Q}{x+1}+\frac{R}{(x-1)^2}$

40. A particle is executing simple harmonic motion with amplitude 2 meters and period 12 seconds. What is the maximum speed in m/s of the particle?

A $\frac{\pi}{6}$

B $\frac{\pi}{3}$

C $\frac{\pi^2}{36}$

D $\frac{\pi^2}{3}$

41. Which one the following linear Diophantine equations has positive solutions?

A $x+5y=7$

B $x+5y=3$

C $x+5y=2$

D $x+5y=1$

42. A force $\mathbf{F}=(2\mathbf{i}+4\mathbf{j}-\mathbf{k})N$, displaces a particle through $(12\mathbf{i}+4\mathbf{j}+2\mathbf{k})m$. The work done by the force is

A 23 J

B 38 J

C 48 J

D $\sqrt{21}\times\sqrt{164}$ J

43. If $|x|>2$, then $\int \frac{1}{\sqrt{x^2-4}}dx=$

A $\frac{1}{2}\cos^{-1}\left(\frac{x}{2}\right)+k$

B $\sin^{-1}\left(\frac{x}{2}\right)+k$

C $\cosh^{-1}\left(\frac{x}{2}\right)+k$

D $\frac{1}{2}\tanh^{-1}\left(\frac{x}{2}\right)+k$

44. The furthest point from the pole to the polar curve $r=2+\sqrt{2}\sin\theta$ occurs when $\theta=$

A $-\frac{\pi}{2}$

B 0

C $\frac{\pi}{2}$

D π

45. A point P traces the curve $r=a\theta$, O the pole such that OP moves with constant angular velocity w. The radial component of acceleration when P is at the point (r,θ) is

A $-rw^2$

B rw^2

C $-2aw^2$

D $2aw^2$

46. If $f(x)=2x+|x-2|-|x|$ and $0\le x<2$, then $f(x)=$

A 2

B $2+4x$

C $2-4x$

D $-2+2x$

47. The component of the force $(3\mathbf{i}-\mathbf{j}+2\mathbf{k})N$ in the direction the vector $(-4\mathbf{j}+3\mathbf{k})$ is

A 10

B 9

C 9/5

D 2

48. The area of a square, $A(t)$, is increasing at a rate equal to its perimeter. $A(t)$ satisfies the differential equation $\frac{dA}{dt}=$

A $4A$

B $2A$

C $-4\sqrt{A}$

D $4\sqrt{A}$

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49. Given that the moment of inertia of a rod of mass m and length $2l$ about an axis through its centre perpendicular to the rod is I, then the moment of inertia about a perpendicular axis on the rod distant x from the centre is

A $I - mx^2$

B $I + mx^2$

C $\frac{4}{3}ml^2 + mx^2$

D $\frac{1}{3}I + mx^2$

50. How many proper subgroups are there in a cyclic group of order 12?

A 4

B 3

C 2

D 1

STOP

GO BACK AND CHECK YOUR WORK

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CAMEROON GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2020 **ADVANCED LEVEL**

Subject Title	**Further Mathematics**
Paper No.	**Paper 2**
Subject Code No.	**0775**

THREE Hours

Answer ALL 10 questions.

For your guidance, the approximate mark allocation for parts of each question is indicated.

Mathematical formulae and tables published by the Board, and noiseless non-programmable electronic calculators are allowed.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

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1. Use the substitution $y = vx$, where v is a function of x, to transform the differential equation

$$x^2\frac{dy}{dx} = x^2 + xy + y^2$$

into a differential equation in v and x. **(3 marks)**

Hence find the solution of this differential equation given that $y = 1$ when $x = 1$. **(5 marks)**

2. Solve for real x, the equation

$$3\cosh 2x = 3 + \sinh 2x .$$ **(6 marks)**

3. Given that $$I_n = \int_0^1 x(1-x)^n\,dx ,$$

where n is a positive integer,
show that
$(n+2)I_n = nI_{n-1}$, $n \geq 1$. **(5 marks)**

Hence, evaluate $\int_0^1 x(1-x)^5\,dx$. **(3 marks)**

4. Find the cartesian equation of the curve with polar equation

$$r = \cos^2\theta .$$ **(3 marks)**

Hence, show that the equation of the tangent to the curve at the point $\left(\frac{1}{2}, \frac{\pi}{4}\right)$ is

$$r = \frac{1}{\sqrt{2}(3\sin\theta - \cos\theta)}$$ **(4 marks)**

5. Show that the equation of the normal to the hyperbola $xy = 4$ at the point $P\left(2t, \frac{2}{t}\right)$, $t \neq 0$ is

$$t^3x - ty - 2t^4 + 2 = 0 .$$ **(3 marks)**

This normal cuts the hyperbola again at the point Q.

(i) Find the coordinates of Q. **(2 marks)**

(ii) Prove that the locus of the midpoint of the origin and Q is another hyperbola. **(3 marks)**

6. Given that $z = \cos\theta + i\sin\theta$,

(i) Show that

$$z + \frac{1}{z} = 2\cos\theta .$$ **(2 marks)**

(ii) Find in a similar manner, an expression for $z^2 + \frac{1}{z^2}$. **(1 mark)**

(iii) Show that

$$z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2} = 4\cos^2\theta - 2\cos\theta .$$ **(2 marks)**

Hence, solve the equation

$$z^4 - z^3 + 2z^2 - z + 1 = 0,$$

giving the roots in the form $a + bi$. **(3 marks)**

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6. A particle P moves on the curve with polar equation

$$r = \frac{1}{2 - \sin\theta}.$$

Given that at any instant t, during the motion,

$$r^2 \frac{d\theta}{dt} = 4,$$

(i) write an expression for $r\frac{d\theta}{dt}$ in terms of θ. **(2 marks)**

(ii) show that

$$\frac{dr}{dt} = 4\cos\theta \text{ and } \tfrac{1}{3} \le r \le 1.$$ **(3 marks)**

(iii) Find the speed of P when $\theta = 0$. **(4 marks)**

(iv) Prove that the force acting on P is directed towards the pole. **(4 marks)**

---

7. Three vectors

$$\mathbf{a} = (3\mathbf{i} + 4\mathbf{j}),\ \mathbf{b} = (-4\mathbf{i} + 3\mathbf{j}),\ \mathbf{c} = (-3\mathbf{i} + 4\mathbf{j})$$

lie on a horizontal plane, Π.

(i) Determine which of the vectors $\mathbf{b}$ or $\mathbf{c}$ is perpendicular to $\mathbf{a}$. **(2 marks)**

Two smooth spheres, S and T, each of mass m lie on the plane Π. Sphere S is projected along the plane towards sphere T with velocity $5u(\mathbf{i} + 2\mathbf{j})$ so that it collides obliquely with T. At the instant of collision, the line of impact is parallel to the vector $\mathbf{a}$. Given that the coefficient of restitution between the two spheres is $\frac{1}{2}$.

Show that

(ii) the component of the initial velocity of S parallel to $\mathbf{a}$ is $11u$. **(2 marks)**

(iii) the component of the final velocity of S parallel to $\mathbf{a}$ is $\frac{11}{4}u$. **(4 marks)**

(iv) the magnitude of the impulse exerted by sphere T on sphere S is $\frac{33}{4}mu$. **(4 marks)**

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8. The lengths of rods from a certain factory are normally distributed with mean μ cm and standard deviation 6cm. It is known that 4.78% of the rods have lengths greater than 82cm.

Find to the nearest integer,

(i) the mean μ. **(5 marks)**

(ii) the range, α (symmetrical about the mean) within which 75% of the lengths of the rods lie. **(6 marks)**

Find also, to 4 decimal places,

(iii) the probability that a rod chosen at random has a length between 62cm and 72cm. **(3 marks)**

END

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(b) Given that $g(x) = \dfrac{x^2}{2x-1}$, defined on $]1, +\infty[$, and the sequence (u_n) defined recursively by

$$\begin{cases} u_0 = 2 \\ u_{n+1} = g(u_n) \end{cases}.$$

(i) Show that $\forall x > 1$, $g(x) > 1$. **(3 marks)**

Given the sequences (v_n) and (w_n) such that

$$v_n = \frac{u_n - 1}{u_n} \text{ and } w_n = \ln v_n ,$$

show that

(ii) w_n is a geometric sequence. **(4 marks)**

(iii) $v_n = 2^{-2^n}$. **(2 marks)**

END

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CAMEROON GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2020 **ADVANCED LEVEL**

Subject Title	**Further Mathematics**
Paper No.	**Paper 3**
Subject Code No.	**0775**

Two and a half hours

Answer ALL questions.

For your guidance the approximate mark allocation for parts of each question is indicated in brackets.

Mathematical formulae and tables, published by the Board, and noiseless non-programmable electronic calculators are allowed.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

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1. A particle P, of mass m, is projected vertically upwards from a point O with speed $\frac{2g}{k}$ ms^{-1} in a medium whose resistance to motion is of magnitude kv per unit mass, where v is the speed of P and k a positive constant.

(i) Show that

$$v\frac{dv}{dx} = -g - kv$$ **(1 mark)**

The particle P attains the highest point H of its path.

(ii) Show further that

$$\text{OH} = \frac{g}{k^2}(2 - \ln 3) \text{ metres}$$ **(5 marks)**

Given that the particle P reaches another point M with speed $\frac{g}{2k}$ ms^{-1} as it traces its path back to O, the point of projection,

(iii) Prove that

$$\text{HM} = \frac{g}{k^2}(\ln 2 - \tfrac{1}{2})$$ **(4 marks)**

---

2. Given that y satisfies the differential equation

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} + xy = 0 \text{ and that } y = 1, \frac{dy}{dx} = 0 \text{ when } x = 0.$$

Using the approximations

$$h^2\left(\frac{d^2y}{dx^2}\right)_n \cong y_{n+1} - 2y_n + y_{n-1} \text{ and } 2h\left(\frac{dy}{dx}\right)_n \cong y_{n+1} - y_{n-1}$$

and a step-length of 0.2,

show that

(i) $y_{n+1} \cong \dfrac{2x_n y_n(2 - h^2) + y_{n-1}(h - 2x_n)}{h + 2x_n}$, **(2 marks)**

(ii) $y_{-1} = y_1 = 1$. **(4 marks)**

Hence, find correct to 4 decimal places,

(iii) the value of y when $x = 0.6$. **(6 marks)**

3. A particle A moves in a resisting medium in a straight line such that its distance x from a fixed point O satisfies the equation

$$\frac{d^2x}{dt^2} + p\frac{dx}{dt} + qx = 0, \text{ where } p \text{ and } q \text{ are constants.}$$

Find the condition(s) on p and q such that the motion of A is

(i) simple harmonic. **(3 marks)**

(ii) damped harmonic. **(4 marks)**

In the case where the motion is damped harmonic, find

(iii) the damping factor. **(3 marks)**

(iv) the period of the motion. **(2 marks)**

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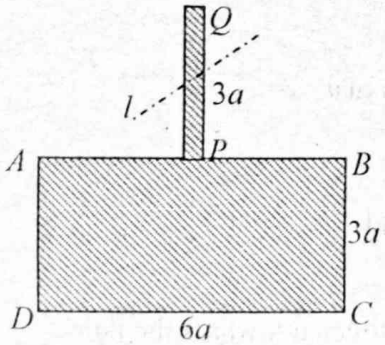
4. A plane composite object is made by attaching a thin uniform rod PQ of mass M, at right angles to the midpoint AB of a uniform rectangular lamina $ABCD$ of mass m. Given that

$$AB = 6a,\ BC = 3a = PQ,$$

as shown in the diagram below:



(i) Show that the moment of inertia I of the object about a smooth horizontal axis l, through the midpoint of PQ and perpendicular to the plane of the lamina, is given by

$$I = \tfrac{3}{4}\left(17m + M\right)a^2$$ **(3 marks)**

The object is slightly disturbed through small angles θ from its equilibrium position to swing below Q. After time t,

(ii) show that

$$I\frac{d^2\theta}{dt^2} = -3mga\sin\theta\,.$$ **(4 marks)**

(iii) find an expression for the period of small oscillations of the object about l. **(3 marks)**

(iv) show further that if $m = M$, then the component of the force perpendicular to the rod exerted by the axis of rotation is $\frac{8}{3}mg\sin\theta$. **(4 marks)**

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5. A system of forces $\mathbf{F}_1, \mathbf{F}_2$ and $\mathbf{F}_3$ acting through points with position vectors $\mathbf{r}_1, \mathbf{r}_2$ and $\mathbf{r}_3$ respectively, where

$$\mathbf{F}_1 = (3\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})N, \qquad \mathbf{r}_1 = (\mathbf{i} + \mathbf{j} + \mathbf{k})m$$

$$\mathbf{F}_2 = (3\mathbf{i} + 4\mathbf{j} + 3\mathbf{k})N, \qquad \mathbf{r}_2 = (3\mathbf{i} + 2\mathbf{j} + \mathbf{k})m$$

$$\mathbf{F}_3 = (-4\mathbf{i} - 2\mathbf{j} + \lambda\mathbf{k})N, \qquad \mathbf{r}_3 = (2\mathbf{i} - \mathbf{j})m$$

reduces to a couple $\mathbf{G}$ and a single force $\mathbf{F} = (2\mathbf{i} - \mathbf{j} + 8\mathbf{k})N$ acting through the point with position vector

$$\mathbf{r}_0 = (2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

Find

(i) the value of λ, **(2 marks)**

(ii) the vector moment of $\mathbf{F}_2$ about the point with position vector $\mathbf{r}_0$, **(3 marks)**

(iii) the Cartesian equation of the line of action of $\mathbf{F}$, **(4 marks)**

(iv) the magnitude of the couple $\mathbf{G}$. **(4 marks)**

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7. (a) Given that S is a similarity transformation(similitude) defined by

$$z' = (1+i)z - 2 + 3i \ ,$$

find

(i) the scale factor **(1 mark)**

(ii) the invariant point (centre) and **(2 marks)**

(iii) the angle of rotation of this transformation. **(2 marks)**

(b) A linear transformation, T, is defined as follows

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$(x, y) \mapsto (2x + y,\ x - y).$$

Determine the kernel of T. **(3 marks)**

8. Solve in $\mathbb{N}$, the system of equations

$$3x \equiv 1 \pmod 5$$
$$5x \equiv 2 \pmod 7$$

(8 marks)

9. A sequence, (u_n), is defined recursively by

$$u_0 = 1\ ,\ u_1 = 3 \text{ and } u_{n+1} = 3u_n + 4u_{n-1}\ .$$

(i) Find u_2. **(2 marks)**

(ii) Show that $u_{n+1} - 4u_n = (-1)^{n+1}(u_1 - 4u_0)$. **(3 marks)**

(iii) Hence express u_{n+1} in terms of u_n. **(1 mark)**

Show that

(iv) u_n is divergent. **(2 marks)**

(v) $u_n \geq 1,\ \forall n \in \mathbb{N}$. **(3 marks)**

(vi) u_n is increasing. **(2 marks)**

Two other sequences (v_n) and (w_n) are defined as

$$v_n = u_n + u_{n-1} \text{ and } w_n = u_n - 4u_{n-1}, n \geq 1.$$

(vii) Show that v_n and w_n are geometric sequences. **(3 marks)**

(viii) Express v_n and w_n each in terms of n. **(2 marks)**

Hence, or otherwise, express u_n in terms of n. **(2 marks)**

10. (a) A function, f, is defined by

$$f(x) = \sqrt{x^2 - 3x + 2}\ .$$

(i) Find the domain of f. **(3 marks)**

(ii) Show that

$$f(x) = \sqrt{\left(x - \frac{3}{2}\right)^2 - \frac{1}{4}} \text{ and that as } x \to +\infty\ ,\ f(x) \to x - \frac{3}{2}.$$

Find a similar expression for $f(x)$ as $x \to -\infty$. **(4 marks)**

(iii) Hence, state the oblique asymptotes of f. **(1 mark)**

(iv) Sketch the graph of $y = f(x)$, showing clearly the asymptotes. **(3 marks)**

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